13 September 2026

Editors
European Journal of Agronomy

Dear Editors,

Please consider our manuscript, “Beyond thermal accumulation: improving cross-regional grapevine phenology prediction with terrain-aware forcing and residual learning”, by Linjia Yao, Qiang Yu and Gang Zhao, for publication as a research article in European Journal of Agronomy.

Reliable phenology estimates connect crop development with the seasonal conditions relevant to vineyard monitoring and management. This study examines a practical model-selection question: whether improved temperature inputs and machine-learning residual correction provide additional value when combined, and how their contributions depend on process structure and developmental stage. The focus on crop development, environmental variation and the agronomic interpretation of model performance aligns with the journal’s interests in crop physiology and modelling.

The analysis includes 74,618 grapevine observations from 3,169 monitoring sites in 11 European countries, covering budburst, flowering, veraison and maturity/harvest. Ten process structures are crossed with raw or elevation-corrected temperature and the absence or presence of residual learning, giving 40 configurations per stage. Calibration and testing keep overlapping locations and shared weather cells together, and residual learners use nested, out-of-fold process predictions. Paired correction comparisons use the same 15,105 observations at 633 test sites.

Elevation correction with threshold recalibration reduces mean absolute error by 1.06 days (95% group-bootstrap interval: 0.70–1.39 days). Residual learning under raw forcing gives a larger reduction of 4.10 days (3.56–4.63 days), whereas adding elevation correction to the hybrids yields only 0.12 days (−0.10 to 0.31 days). These results show that the gains from the two corrections are not additive. They support joint evaluation of weather inputs and residual learning, with final stage-specific error and developmental attainment informing model selection. Veraison and maturity/harvest remain more difficult to predict than budburst and flowering.

The estimates concern retrospective date prediction; operational forecasting and downstream management outcomes were not evaluated. Late or unattained veraison and maturity receive a fixed 4 November date for scoring only. Screening and calibration changed during model development, and independent confirmation is needed before extending the findings to unobserved regions or climates.

Thank you for considering this manuscript.

Yours sincerely,

Gang Zhao
Corresponding author
State Key Laboratory of Soil and Water Conservation and Desertification Control
College of Soil and Water Conservation Science and Engineering
Northwest A&F University, Yangling, Shaanxi 712100, China
gang.zhao@nwafu.edu.cn